MTH 446: Lecture # 1

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August 26, 2026

Lecture Span

1. Introduction into class
2. Introduction into Complex numbers

Introduction

1. 2 Midterms (100 points each); 1st is early october. 2nd is mid november.
2. 1 Final (200 points)
3. Homeworks (60-80 points)
4. Textbook: Churchill & Brown (homeworks come from this)

Complex numbers

Complex numbers are created from the real number line, such that

$$z = a + bi, \quad a, b \in \mathbb{R},\ i \in \mathbb{C} \tag{1}$$

It follows that

$$(a + bi) + (c + di) = (a + c) + (b + d)i \tag{2}$$

Moreover,

$$(a + bi)(c + di) = ac + adi + bci - bd \tag{3}$$

This is just the standard distributive rule. You also have to introduce the complex conjugate:

$$\overline{z} = a - bi, \quad z = a + bi \tag{4}$$

The conjugate is $\overline{z}$. Here, then some more properties:

$$\overline{(z \times w)} = \overline{z} \times \overline{w} \tag{5}$$

$$\overline{z + w} = \overline{z} + \overline{w} \tag{6}$$

Also,

$$z \cdot \overline{z} = a^2 + b^2 \geq 0 \tag{7}$$

Then define

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2} \tag{8}$$

Also

$$z \cdot \frac{z}{|z|^2} = 1 \tag{9}$$

Therefore,

$$z \neq 0 \implies \frac{1}{z} = \frac{\overline{z}}{|z|^2} \tag{10}$$

This means that all non-zero complex numbers have an inverse. Also,

$$|zw| = |z||w| \tag{11}$$

Then, we can also think of such complex numbers geometrically, on a 2-d plane. Consider

$$\frac{z}{|z|} = \frac{a}{\sqrt{a^2+b^2}} + \frac{b}{\sqrt{a^2+b^2}} i \tag{12}$$

This is simiilar to seeing a cosine and a sine. In particular,

$$\cos\theta = \frac{a}{\sqrt{a^2+b^2}}, \quad \sin\theta = \frac{b}{\sqrt{a^2+b^2}} \tag{13}$$

Then, θ is the argument of z, denoted as $\arg(z)$. Then
Is this correct?:

$$z = \frac{z}{|z|}(\cos\theta + i\sin\theta) \tag{14}$$

Moreover,

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{15}$$

This is also called $\text{cis}(\theta)$. Then consider

$$\begin{align}
e^{i\theta} \cdot e^{i\phi} &= (\cos\theta + i\sin\theta)(\cos\phi + i\sin\phi) \tag{16}\\
&= \cos\theta\cos\phi - \sin\theta\sin\phi + (\cos\theta\sin\phi + \sin\theta\cos\phi)i \tag{17}\\
&= \cos(\theta+\phi) + \sin(\theta+\phi)i \tag{18}\\
&= e^{i(\theta+\phi)} \iff \text{cis}(\theta+\phi) \tag{19}
\end{align}$$

You can also derive $e^{i\theta}$ from the taylor series expansion. I won't add that here.
Also, you can think of z as

$$z = |z|e^{i\theta} \tag{20}$$

Towards next class, a complex function is a function of two variables. Such that

$$f(x+iy) = u(x,y) + iw(x,y) \tag{21}$$

Then define a complex derivative

$$\frac{f(z+\Delta z)}{\Delta z} \tag{22}$$